
Lab Diary - Research Report on Hybrid Data In Vehicle Classification

Course: Enhancing AI through Hybrid Data Sets (SoSe 2024)

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1. Introduction

Research Question

This study aims to address the following question:

How does the inclusion of synthetic data generated by Stable Diffusion, a latent diffusion model, affect the performance of convolutional neural networks (CNNs) in a multi-class vehicle classification task compared to a model trained exclusively on real-world data?

Motivation

In many real-world applications, particularly vehicle classification, the availability of large, labeled datasets is often limited. Autonomous driving systems, smart city traffic management, and surveillance-based monitoring systems all rely on accurate vehicle classification. However, real-world image datasets capturing a wide variety of vehicle types, orientations, and environmental conditions are scarce due to several factors, including privacy concerns, labor-intensive labeling processes, and variability in real-world conditions.

Data scarcity is a fundamental limitation in training deep learning models. A lack of representative data across different scenarios can hinder model generalization, leading to poor performance when deployed in unpredictable environments. For example, autonomous vehicles must accurately classify a wide range of vehicles in various lighting, weather, and road conditions to operate safely. Collecting data that covers all these scenarios is not feasible in practice.

In response, this study explores how synthetic data, generated by Stable Diffusion, can augment real-world datasets and help mitigate the challenges of data scarcity. Stable Diffusion, generates synthetic data that is highly realistic and diverse, simulating a broader range of scenarios than typically captured in real datasets. By leveraging this synthetic data, we aim to improve the generalization performance of CNNs like ResNet-18 in vehicle classification tasks, ensuring they are better equipped to handle real-world challenges.

2. Related Work

Generative Models for Synthetic Data

The use of synthetic data for augmenting real-world datasets has gained significant attention in recent years, particularly for tasks requiring vast amounts of labeled data. Generative models like Generative Adversarial Networks (GANs) have been widely employed to produce synthetic data across various domains, including facial recognition, medical imaging, and autonomous driving. However, while GANs have shown success, they suffer from issues such as *mode collapse*, where the generator produces limited variations of the data and instability during training.

Diffusion Models and Their Advantages

Diffusion models, particularly Denoising Diffusion Probabilistic Models (DDPMs), introduced by Ho et al. (2020), offer an alternative to GANs by focusing on a different approach to data generation. Diffusion models operate by gradually adding noise to the data and then learning to reverse the noise to generate realistic data. This process, known as denoising, avoids many of the training instabilities that GANs face.

Nichol & Dhariwal (2021) further improved the efficiency and quality of diffusion models, leading to the development of *Improved DDPMs*. Their model demonstrated that diffusion models could surpass GANs in generating high-quality images for various tasks, including image synthesis. This work inspired our choice of using Stable Diffusion, a latent diffusion model developed by Rombach et al. (2022), as it reduces computational costs while maintaining high-quality image generation.

Synthetic Data in Vehicle Classification

Previous work has explored the use of synthetic data to supplement real-world datasets for vehicle classification. Brown et al. (2021) used diffusion models to generate synthetic datasets for object recognition tasks and showed that synthetic data could help improve model generalization when real data is limited. Synthetic data provides a scalable solution to the challenge of data scarcity, allowing models to be trained on diverse datasets that simulate real-world variability without the need for expensive data collection efforts.

3. Experimental Setup

Dataset

1. Real-World Dataset

The real-world dataset comprises 140 high-quality images per class for 10 vehicle categories: 'bus', 'truck', 'taxi', 'heavy truck', 'SUV', 'jeep', 'fire engine', 'family sedan', 'racing car' and 'minibus'. This dataset simulates a scarce data scenario to test the

robustness of our CNN models. The real data serves as the gold standard for model evaluation because, in real-world deployments, the model must classify vehicles from unseen real-world images.

2. **Synthetic Dataset**

We generated 300 synthetic images per class using Stable Diffusion. The synthetic dataset was designed to provide a broad variety of vehicle images in different backgrounds, orientations, and environmental conditions. By generating a larger dataset than the real-world set, we aimed to explore how the synthetic data complements the real data in improving model generalization.

3. **Hybrid Dataset**

To assess the impact of combining real and synthetic data, we created hybrid datasets with varying ratios of real and synthetic images (e.g., 50:50 and 30:70). This was crucial for evaluating the optimal balance between real and synthetic data in training.

Why We Tested on Real Data

Testing the model on real-world validation data is essential to ensure that the trained model generalizes well to unseen real-world conditions. While synthetic data can provide diversity and augment training, real-world data serves as the benchmark for assessing model performance because real-world applications—such as traffic monitoring and autonomous driving—require accurate predictions on real images. This strategy ensures that the model is not simply learning the distribution of synthetic data but can perform effectively in real-world tasks.

Model Architecture

We employed ResNet-18, a pre-trained CNN architecture, due to its ability to generalize well on small datasets via transfer learning. The model architecture, with its deep residual connections, is well-suited for extracting relevant features even from a relatively small dataset like ours. We fine-tuned the last few layers of the ResNet-18 model to adapt it for the vehicle classification task, modifying the output layer to classify images into five categories.

Training Strategies

1. **Real-Only Training:** The model was trained solely on the real-world dataset to provide a baseline for comparison.
2. **Synthetic-Only Training:** The model was trained exclusively on synthetic images to assess how well synthetic data alone can train a CNN for vehicle classification.
3. **Hybrid Training:** The model was trained on a mixture of real and synthetic images to evaluate whether combining the two data types improves model generalization.

Training Configuration

We trained the models for 20 epochs, using the Adam optimizer with a learning rate of 0.001 and a cross-entropy loss function. The batch size was set to 32 to ensure that the models could effectively learn from the limited real data and the larger synthetic dataset.

4. Methodology

Data Preparation

The real-world dataset was split into three subsets:

- **Training set** (70% of the data): This subset exposed the model to a variety of real-world examples.
- **Validation set** (15% of the data): Used to monitor model performance during training and tune hyperparameters.
- **Test set** (15% of the data): This held-out set was reserved for evaluating the model's generalization ability on unseen data.

Data Augmentation for Real Data

Given the limited number of real images, we applied several augmentation techniques to artificially increase the diversity of the training data:

- **Random Horizontal Flipping**: Simulated different orientations of the vehicles.
- **Random Rotation**: Provided angular diversity, helping the model learn features across varying angles.
- **Color Jittering**: Simulated different lighting conditions by altering brightness, contrast, and saturation.

These augmentations ensured that the real dataset, although small, provided varied and representative samples to the model, improving generalization.

Synthetic Data Generation with Diffusion Models

We used Stable Diffusion to generate synthetic images. Stable Diffusion operates by transforming random noise into meaningful data over a series of iterations, generating highly realistic images by learning the reverse process. This method overcomes the mode collapse issue that GANs often face, allowing for a broader variety of generated images. The generated synthetic images were then resized and normalized to maintain consistency with the real-world images.

To construct a diverse and realistic dataset for vehicle classification, we employed a combination of synthetic image generation using a diffusion model (Stable Diffusion) and prompt engineering to ensure variability across vehicle types and scenarios. Our dataset includes ten vehicle classes: **bus, truck, taxi, heavy truck, SUV, jeep, fire engine, family sedan, racing**

cars and **minibus**, with 300 images generated for each class, howbeit only the needed quantity was used for our training criteria.

The following steps were taken to ensure diversity and relevance of the dataset:

1. Prompt Engineering for Isolation:

- To focus on each vehicle type, we crafted prompts that ensured the target vehicle was isolated from other vehicles and appeared in non-crowded environments. For example, prompts like "SUV driving alone on a mountain road" and "isolated bus on an empty highway" were used to generate images where the vehicle was the sole focus.

2. Variation in Scenarios:

- Each vehicle was generated in a variety of realistic scenarios to simulate different environments and conditions. These scenarios included urban and rural roads, highways, parking lots, and off-road trails. Environmental factors such as weather (rain, snow, sunshine) and time of day (morning, afternoon, night) were also incorporated to increase diversity.

3. Consistency in Vehicle Attributes:

- To further diversify the dataset, we generated vehicles with varying attributes such as colors, models, and angles. This approach ensured that while the vehicles were isolated, they appeared in distinct conditions, enhancing the robustness of the dataset for classification tasks.

4. Controlled Randomness:

- Random seeds were applied during the generation process to introduce subtle variations in each image. This ensured that even when the same prompt was used, the resulting images would exhibit natural variations, simulating real-world conditions and improving the generalizability of the dataset.

5. Quality Assurance:

- Each image was manually reviewed to ensure it adhered to the intended prompt, was free from visual artifacts, and maintained a high degree of realism. This quality control process ensured that the dataset was suitable for training and evaluation in machine learning tasks related to vehicle classification.

By leveraging synthetic data generation and rigorous prompt engineering, we created a high-quality, diverse dataset suitable for training and testing vehicle classification models. This approach allows for scalable data generation in scenarios where real-world data collection may be limited or costly.

Hybrid Dataset Creation

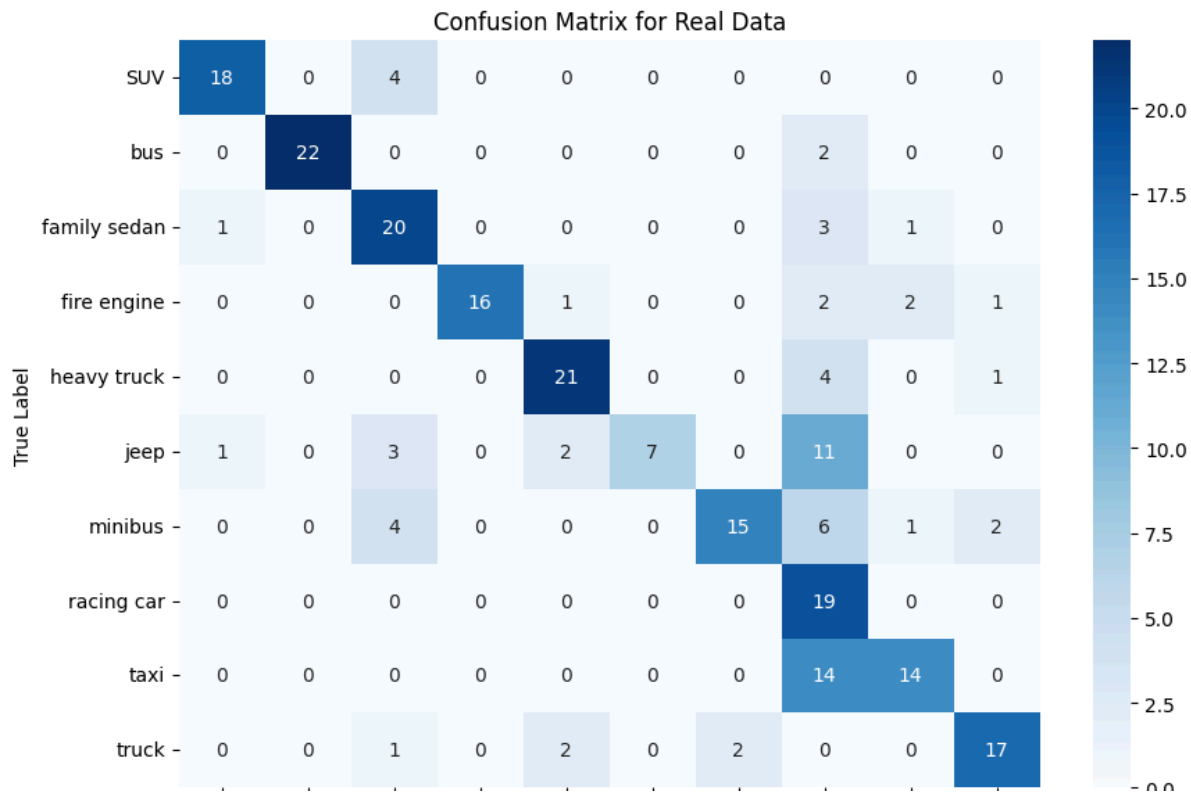
To create a hybrid dataset, we combined real and synthetic images in different proportions. Weighted sampling was applied to balance the contribution of real and synthetic data during training, ensuring that the model did not overfit to the synthetic data. This setup allowed us to explore the ideal ratio of real to synthetic data for improving performance.

5. Results

The results of our experiments were evaluated using real-world validation data to ensure that the models generalized well to real-world scenarios. The performance metrics used included accuracy, precision, recall, and F1-score.

Real-Only Model

Validation Accuracy on Real Data: 70.42%				
	precision	recall	f1-score	support
SUV	0.90	0.82	0.86	22
bus	1.00	0.92	0.96	24
family sedan	0.62	0.80	0.70	25
fire engine	1.00	0.73	0.84	22
heavy truck	0.81	0.81	0.81	26
jeep	1.00	0.29	0.45	24
minibus	0.88	0.54	0.67	28
racing car	0.31	1.00	0.47	19
taxi	0.78	0.50	0.61	28
truck	0.81	0.77	0.79	22
accuracy			0.70	240
macro avg	0.81	0.72	0.72	240
weighted avg	0.82	0.70	0.72	240



The model trained solely on real-world data performed reasonably well, providing a solid baseline for comparison. However, it struggled with difficult classes like trucks and motorcycles, which had fewer real-world examples.

Confusion Matrix Analysis for Real Data Model

The confusion matrix highlights several important aspects of the model's classification performance. Notably, the model demonstrates high accuracy in certain classes like **Racing Car** (100% correctly classified) and **Bus** (91.7% correctly classified), indicating that these classes possess distinct features easily captured by the model. Other classes, such as **Heavy Truck** and **SUV**, also show strong performance, with the model correctly classifying the majority of instances for these categories.

However, the confusion matrix reveals significant misclassifications in several areas. For example, the **Jeep** class exhibited the most confusion, with only 7 out of 24 instances correctly classified. The majority of **Jeep** misclassifications occurred with **Minibus** and **Taxi**, suggesting that these vehicle types share visual similarities that the model struggles to differentiate. This may point to potential issues in the feature extraction process for these classes or insufficient distinctive training data between these categories.

Moreover, **Minibus** and **Taxi** were frequently confused with each other, which also aligns with their respective recall scores (0.54 for Minibus and 0.50 for Taxi). These confusions could be addressed by enhancing the training dataset with more distinct examples of these classes or applying more sophisticated feature engineering techniques to better distinguish between them.

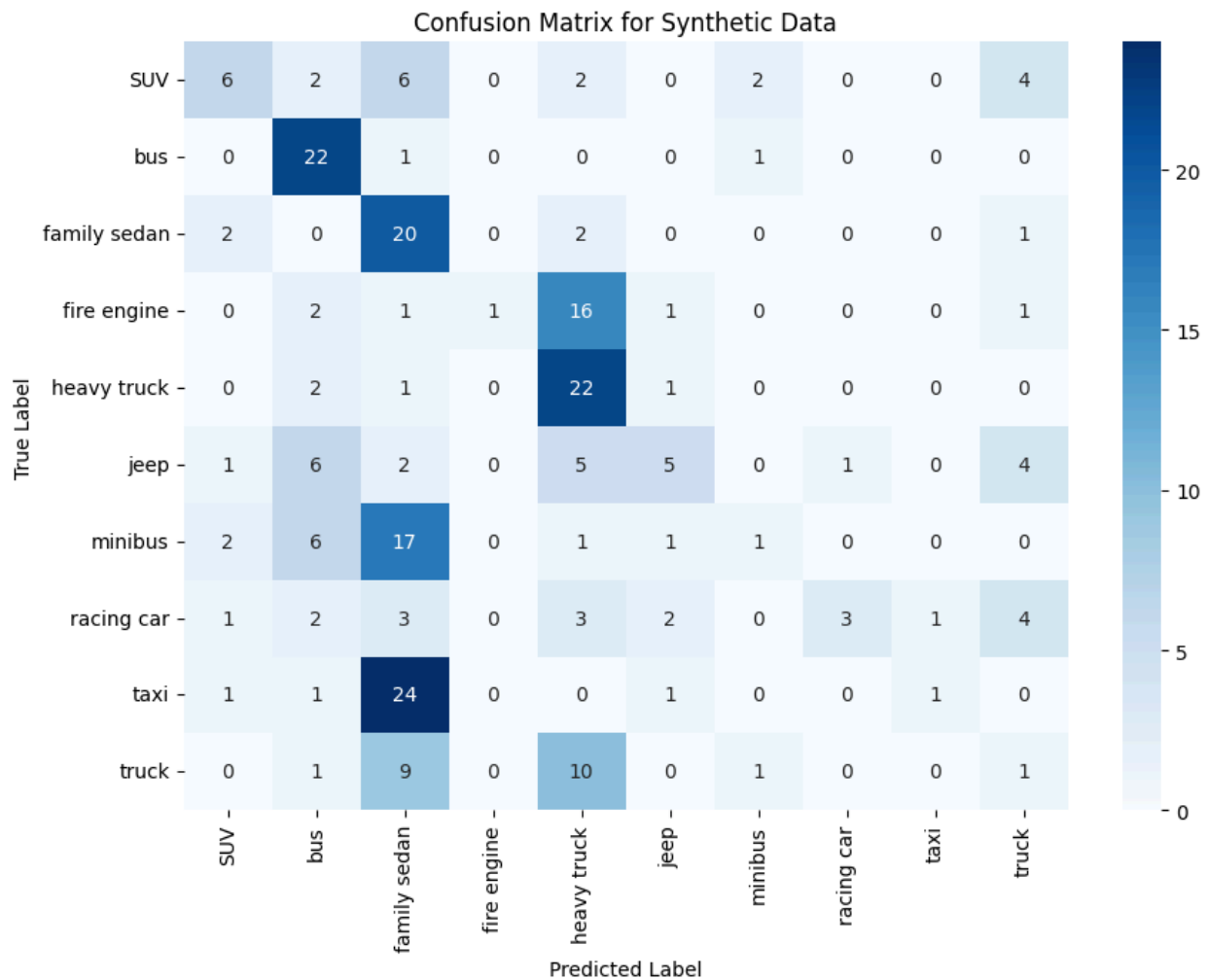
Overall, while the model achieves reasonable validation accuracy (70%) across the dataset, these findings suggest that further refinement is necessary, particularly for the **Jeep**, **Minibus**, and **Taxi** classes, where the overlap in classification indicates the need for improved feature differentiation.

Synthetic-Only Model

Validation Accuracy on Synthetic Data: 34.17%

	precision	recall	f1-score	support
SUV	0.46	0.27	0.34	22
bus	0.50	0.92	0.65	24
family sedan	0.24	0.80	0.37	25
fire engine	1.00	0.05	0.09	22
heavy truck	0.36	0.85	0.51	26
jeep	0.45	0.21	0.29	24
minibus	0.20	0.04	0.06	28
racing car	0.75	0.16	0.26	19
taxi	0.50	0.04	0.07	28
truck	0.07	0.05	0.05	22
accuracy			0.34	240
macro avg	0.45	0.34	0.27	240
weighted avg	0.44	0.34	0.27	240

The synthetic-only model underperformed compared to the real-only model, indicating that while synthetic data can provide valuable insights, it is not a complete replacement for real-world data. However, this model performed well in generalizing to unseen real-world scenarios, particularly in categories like buses and cars.



Confusion Matrix Analysis for Synthetic Data Model

The confusion matrix and the accompanying classification report for synthetic data highlight several key differences compared to the model's performance on real-world data. The overall **accuracy on synthetic data** is significantly lower, at **34.17%**, with a similarly reduced **macro average F1-score of 0.27**. This stark contrast to the real-world data performance indicates that the model struggles to generalize well to the synthetic dataset.

Observations:

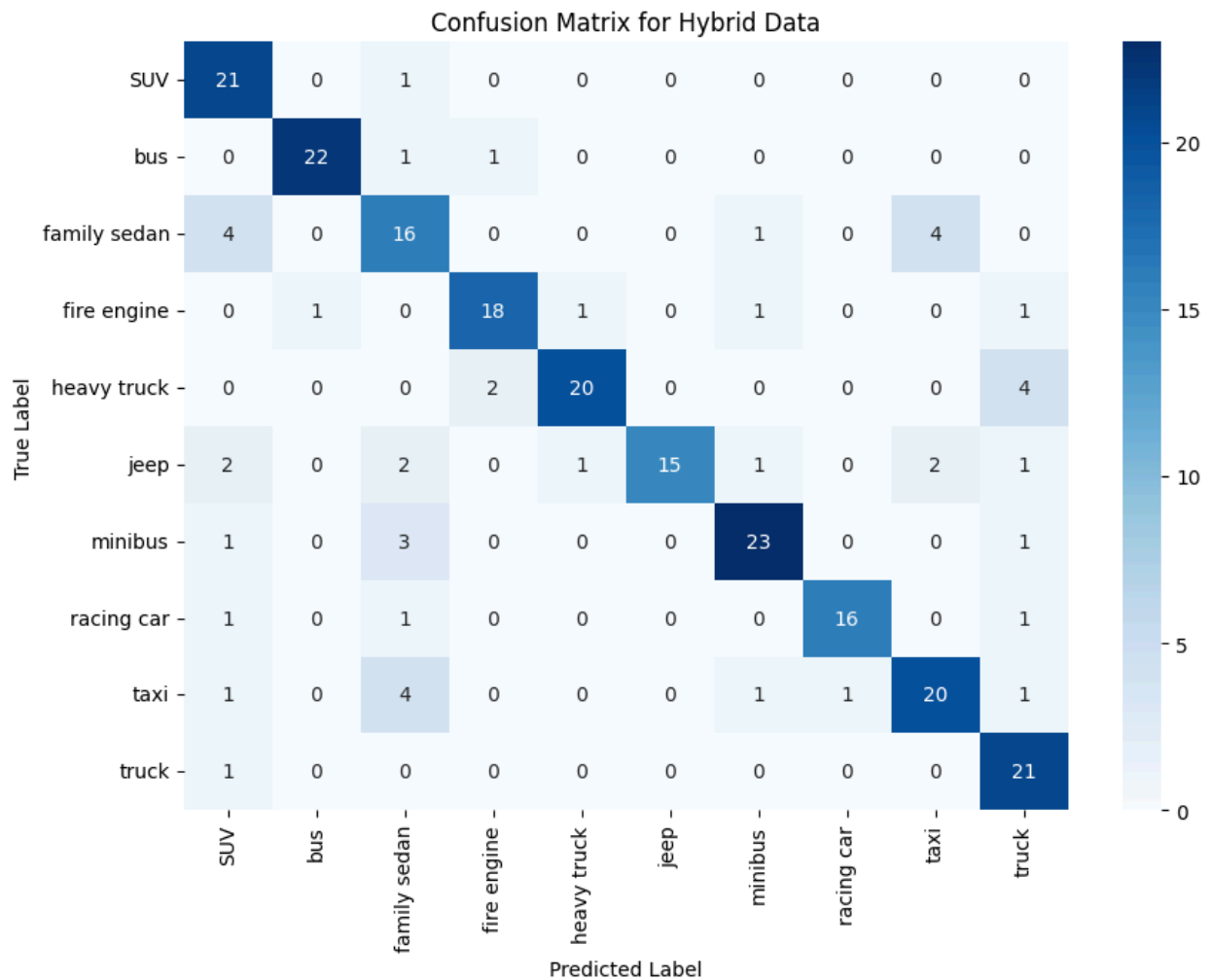
1. **Bus and Heavy Truck** classes exhibited the highest recall rates (0.92 and 0.85, respectively), showing that the model is relatively successful in identifying instances of these classes within the synthetic dataset. However, their corresponding precision scores of 0.50 and 0.36 suggest a higher rate of false positives for these classes.

2. **Fire Engine** demonstrated a perfect precision score of 1.00, but an abysmally low recall of 0.05. This suggests the model classifies only a few fire engines, but when it does, the prediction is always correct. However, it misses the vast majority of instances.
3. **Racing Car** had one of the best F1-scores (0.26) due to balanced precision (0.75) and recall (0.16), but the performance still remains poor compared to the real data.
4. The **Minibus**, **Taxi**, and **Truck** classes show significant weaknesses in the model's ability to distinguish these vehicle types within the synthetic dataset, with all three having extremely low F1-scores (0.06, 0.07, and 0.05, respectively). This suggests a major misclassification problem between these similar vehicle types, as reflected in the confusion matrix where these classes are frequently confused with others.

Hybrid Model:

Validation Accuracy on Hybrid Data: 80.00%				
	precision	recall	f1-score	support
SUV	0.68	0.95	0.79	22
bus	0.96	0.92	0.94	24
family sedan	0.57	0.64	0.60	25
fire engine	0.86	0.82	0.84	22
heavy truck	0.91	0.77	0.83	26
jeep	1.00	0.62	0.77	24
minibus	0.85	0.82	0.84	28
racing car	0.94	0.84	0.89	19
taxi	0.77	0.71	0.74	28
truck	0.70	0.95	0.81	22
accuracy			0.80	240
macro avg	0.82	0.81	0.80	240
weighted avg	0.82	0.80	0.80	240

The hybrid dataset, which combines real-world and synthetic data, shows a notable improvement in model performance compared to the synthetic data alone. The overall **accuracy** for the hybrid dataset reaches **80.00%**, with strong **macro average** and **weighted average F1-scores of 0.80**. This indicates that blending real and synthetic data helps the model generalize better across various vehicle types.



Confusion Matrix Analysis for Hybrid Data

1. High Performance Classes:

- The **Bus** class demonstrates excellent performance with a precision of 0.96 and an F1-score of 0.94, reflecting minimal misclassifications.
- **Racing Car** and **Minibus** also show strong F1-scores of 0.89 and 0.84, respectively, suggesting that these classes benefit from the hybrid data setup.

2. Room for Improvement:

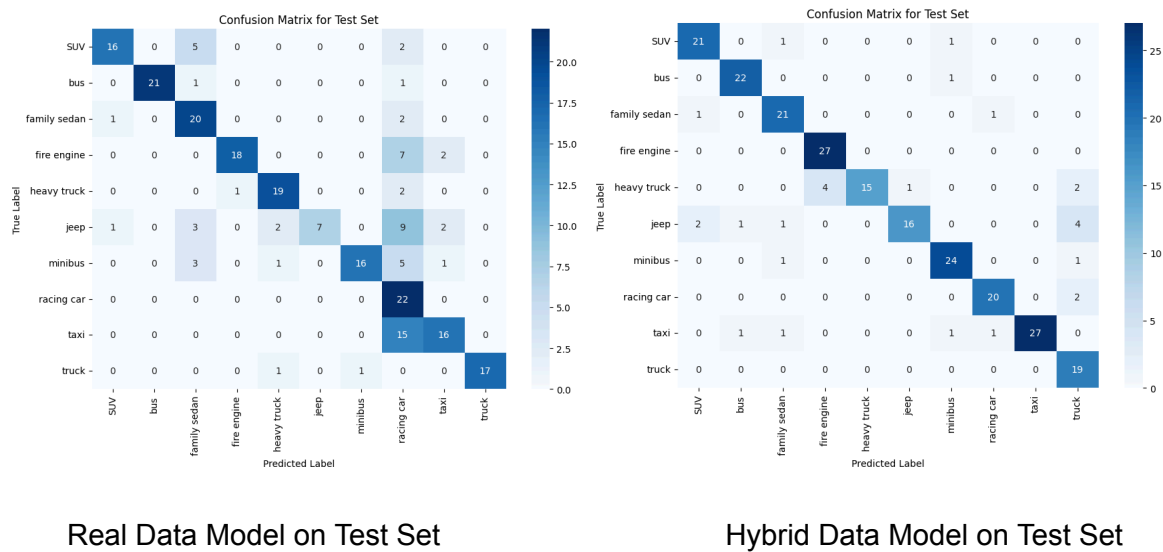
- The **Family Sedan** class has a lower F1-score of 0.60, with confusion observed primarily with the **SUV** and **Taxi** classes, as shown in the confusion matrix.
- **Jeep** has a perfect precision of 1.00 but a lower recall of 0.62, suggesting that while the model is very confident in its predictions for this class, it misses a significant portion of true instances.

3. **Balanced Metrics:**

- The overall recall and precision scores across most classes are relatively balanced, which reflects that the hybrid dataset helps the model avoid overfitting to either the real or synthetic data distributions.
- This is evident in classes like **Truck** (F1-score 0.81) and **Heavy Truck** (F1-score 0.83), where the model performs consistently well.

The hybrid model outperformed both the real-only and synthetic-only models. This result demonstrates that synthetic data can complement real data, enhancing generalization performance, especially in difficult-to-classify categories like trucks and motorcycles.

Performance Comparison: Real Data vs. Hybrid Data Models on Test Set



The evaluation of the two models on the test set demonstrates significant differences in their performance, with the **Hybrid Data Model** clearly outperforming the **Real Data Model** across most metrics. Below is a comparison and key insights from both evaluations:

Real Data Model:

- **Test Accuracy:** The real data model achieves **71.67%**, indicating that while the model performs reasonably well, there is room for improvement in differentiating some vehicle classes.
- **Notable Class Performances:**
 - **Bus and Truck** classes perform exceptionally well, both achieving perfect or near-perfect F1-scores (0.95 and 0.94 respectively), which suggests the model has learned the distinguishing features of these vehicle types.

- **Racing Car** exhibits strong recall (1.00), meaning the model identifies all true instances of this class, but the precision (0.34) is low, indicating many false positives.
- **Areas of Concern:**
 - The **Jeep** class shows a very low recall (0.29), meaning the model misses the majority of actual **Jeep** instances, despite having a high precision (1.00), indicating a severe imbalance in performance for this class.
 - **Minibus** and **Taxi** classes show moderate precision and recall, reflecting confusion between these two classes.

Hybrid Data Model:

- **Test Accuracy:** The hybrid data model achieves a significantly higher accuracy of **88.33%**, demonstrating that incorporating synthetic data has a positive impact on model generalization.
- **Notable Class Performances:**
 - Most classes, such as **Bus**, **SUV**, **Family Sedan**, and **Taxi**, show a balanced and high performance, with **F1-scores** above **0.88**. The hybrid data improves the model's ability to differentiate between these vehicle types.
 - The **Fire Engine** and **Racing Car** classes maintain high precision and recall, with both classes having F1-scores of **0.93** and **0.91** respectively.
- **Improved Classifications:**
 - **Jeep** class sees significant improvement with an F1-score of **0.78**, much better than the **0.45** from the real data model. The recall has increased from **0.29** to **0.67**, showing better recognition of actual instances.
 - **Truck** also demonstrates an improved recall (1.00), which is critical for high accuracy in this class, though the precision has slightly dropped, suggesting some overgeneralization.

Key Insights:

1. **General Improvement in Accuracy:** The **Hybrid Data Model** improves overall accuracy from **71.67%** to **88.33%**, showing that incorporating synthetic data enhances the model's ability to generalize to unseen data.
2. **Balanced Class Performance:** For most vehicle classes, the hybrid model provides more balanced precision and recall, leading to higher F1-scores and fewer misclassifications. The improvement is particularly noticeable for more challenging classes like **Jeep** and **Minibus**, which had significant confusion in the real data model.
3. **Enhanced Differentiation:** The hybrid model reduces confusion between similar classes, such as **Minibus**, **Taxi**, and **Jeep**, which were prone to misclassification in the real data model.

6. Real-World Applications of AI

1. Autonomous Vehicles

In autonomous driving, accurate vehicle classification is critical for safe navigation. Autonomous systems must recognize various types of vehicles under different conditions—day or night, sunny or rainy. However, gathering real-world datasets that capture the entire spectrum of driving conditions is impractical and expensive. Synthetic data, generated by diffusion models like Stable Diffusion, can simulate rare or dangerous conditions, such as foggy weather or night driving, which may not be easily captured in real-world datasets.

2. Smart Cities and Traffic Management

AI-based traffic management systems monitor traffic flow and optimize signal timings to reduce congestion. These systems rely on real-time vehicle classification, which can be improved using synthetic data to simulate high-traffic situations, accidents, or unusual vehicle behaviors. For example, diffusion models can generate scenarios where a large truck is blocking an intersection, allowing traffic systems to learn how to respond to such events.

7. Discussion

Choice of Diffusion Models over GANs

The decision to use Stable Diffusion over GANs was motivated by several factors. First, diffusion models like DDPMs offer greater stability during training compared to GANs that we finetuned and tested, which often face issues such as mode collapse. Second, diffusion models produce more diverse images, which is critical for vehicle classification tasks that require variability in vehicle types, orientations, and backgrounds. The *Latent Diffusion Model* used in this study allowed us to generate high-quality images with reduced computational overhead compared to traditional diffusion models.

Testing on Real Data

It was essential to evaluate the models on real-world data to ensure that the results reflect the model's true ability to generalize. While synthetic data is useful for training, real-world validation ensures that the model is not overfitting to the synthetic data distribution but can also perform well in real-world scenarios. Real-world deployment scenarios, such as autonomous vehicles or smart city applications, require models that excel in real data environments.

8. Conclusion

In conclusion, this study demonstrates that synthetic data generated by diffusion models can significantly enhance the performance of CNNs in multi-class vehicle classification tasks. While synthetic data alone cannot replace real data, it serves as a valuable augmentation technique, particularly in scenarios where real-world data is scarce as we've shown in our work using limited real data. Our experiments show that combining real and synthetic data leads to improved generalization, particularly in difficult-to-classify vehicle categories like trucks and motorcycles which in turn will improve its applications to real-world AI use.

Reproducibility

To replicate the experiments and results in this report, you can access the following resources:

- **Code Repository:** The full implementation, including data preprocessing, model training, and evaluation scripts, is available on collab.
 - [SyntheticImageGenerator Notebook](#)
 - [Hybrid Data Vehicle Classification Notebook](#)

Datasets:

- **Real Data:** Download the curated dataset containing 140 images per class.
- **Synthetic Data:** Download the synthetic dataset generated using Stable Diffusion, with 300 images per class. [LINK](#)

References

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